MATH 4432

T07

Validation and Cross-Validation

Statistical Machine Learning

The prediction question

We observe cases with known answers and use them to fit a prediction rule. The real goal is to predict answers for future cases.

A small error on the cases used for fitting does not tell us how well the rule will work on future cases.

This tutorial builds the answer in stages:

1. Keep some answers out of fitting and calculate their prediction errors.
2. Rotate the held-out cases to use the data more fully.
3. Keep model choice separate from its final assessment.

A complete numerical example connects these stages. The appendix supplies the algebra behind the main formulas.

Background: [1, 2]

One observation and one prediction

Symbol	Meaning	Example
x_i	Predictor for case i	An input value, such as $x_i = -3$
y_i	Observed response for case i	The answer, such as $y_i = 3$
$\hat{y}_i = \hat{f}(x_i)$	Prediction from the fitted rule	If $\hat{f}(x) = 11 + x$, then $\hat{y}_i = 8$

The index i labels a case. A hat means that a quantity comes from fitting data. A predictor is also called a feature or input.

Different cases can have the same predictor value and different responses. A new case does not require a previously unseen x value.

A model family and a fitted model

$$f(x) = a + bx \qquad \Longrightarrow \qquad \hat{f}(x) = 11 + x$$

The model family contains all straight lines. Before fitting, the intercept a and slope b are unknown.

Fitting uses the training cases to choose numbers for a and b. The fitted model above has intercept 11 and slope 1.

The model specification also contains choices we make, such as using a line rather than a quadratic curve. Cross-validation can compare those choices.

In every validation round, the candidate family stays fixed while its coefficients are estimated again from that round’s training data.

Prediction loss and mean squared error

A residual is the observed answer minus its prediction. Squared loss measures the size of this error without allowing positive and negative errors to cancel.

$$e_i = y_i - \hat{y}_i, \qquad L_i = (y_i - \hat{y}_i)^2$$

Observed y	Prediction ŷ	Residual y − ŷ	Squared loss
3	8	−5	25
9	10	−1	1

$$\text{MSE} = \frac{\text{sum of squared losses}}{\text{number of scored cases}} = \frac{25 + 1}{2} = 13$$

MSE is an average squared prediction error. Smaller is better when the cases and scoring rule are the same.

Why training error is an easier target

$$(\hat{a}, \hat{b}) = \underset{a,b}{\operatorname{argmin}} \sum_{i \in T} (y_i - a - bx_i)^2$$

T is the training set. Least squares chooses the coefficients that make this training sum as small as possible. “argmin” means the coefficient values attaining that minimum.

A quadratic family contains every line as the special case c = 0. Its best training error can therefore be no larger than the best line’s training error.

This mathematical advantage concerns fitting the observed answers. Prediction also depends on how well the fitted pattern carries over to other cases.

Validation scores answers that were excluded from the fit. It checks whether the resulting rule predicts beyond its fitting success.

[1, 2]

Training, validation and test roles

Role	What these responses help determine
Training	The coefficients and other fitted quantities
Validation	A choice between candidate model specifications
Test	An assessment after the choices are complete

Training and validation together form the development data. A final test set stays outside model choice.

“Validation” also has a broader meaning: evaluating held-out predictions. In cross-validation, cases rotate through training and held-out roles.

For a candidate specified in advance, its held-out answers must stay outside the steps used to construct their predictions.

[1, 2]

The complete development dataset

There are 12 distinct cases. Each cell below is one response at the indicated predictor value. A, B and C label the folds fixed for this example.

Predictor x	Fold A: y	Fold B: y	Fold C: y
−3	3	9	15
−1	9	7	5
1	11	9	7
3	9	15	21

These hand-constructed data and folds make every calculation checkable. They illustrate the method, rather than evidence that one model always works best.

Four additional cases form a separate test set. Their responses will remain unused until model choice and refitting are complete.

The candidate model specifications

Candidate	Prediction rule before fitting	What fitting estimates
Mean only	a	One intercept a
Line	$a + bx$	Intercept a and slope b
Quadratic	$a + bx + cx^2$	Coefficients a, b and c

All candidates use least squares. We compare the fitted candidates using the same held-out cases and squared loss.

The degree is a setting we choose. The coefficients are quantities the training data determine.

The quadratic family can represent more shapes, but extra flexibility must earn its place by improving predictions on excluded answers.

One holdout split

Cases	Role in this round	Number
Folds B and C	Fit each candidate	8
Fold A	Score predictions after fitting	4

First use only B and C to estimate the coefficients. Then insert the four A predictor values into each fitted rule.

Only after the predictions are fixed do we use the A responses to calculate losses. The A answers do not help estimate these coefficients.

$$\text{MSE}_A = \frac{1}{4} \sum_{i \in A} \left(y_i - \hat{f}_{B+C}(x_i) \right)^2$$

The subscript B+C says which data produced the fitted rule. The score averages over A, the cases that were left out.

The mean-only fit and its held-out errors

$$\hat{a} = \bar{y}_{B+C} = \frac{9 + 7 + 9 + 15 + 15 + 5 + 7 + 21}{8} = 11$$

A constant least-squares fit predicts the training response mean for every x. Appendix A1 explains why.

x in A	Observed y	Prediction	Residual	Squared loss
−3	3	11	−8	64
−1	9	11	−2	4
1	11	11	0	0
3	9	11	−2	4

$$\mathrm{MSE}_A(\text{mean}) = \frac{64 + 4 + 0 + 4}{4} = 18$$

Fitting the line using B and C

Least squares estimates slope from how x and y vary together. The intercept then places the line through the training averages.

$$\hat{b} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad \hat{a} = \bar{y} - \hat{b}\bar{x}$$

For the eight B+C training cases:
The predictor mean is $\bar{x} = 0$ and the response mean is $\bar{y} = 11$.
The sum of x_i^2 is 40 and the sum of $x_i y_i$ is 40.

$$\hat{b} = \frac{40}{40} = 1, \qquad \hat{a} = 11 - 1(0) = 11, \qquad \hat{f}_{B+C}(x) = 11 + x$$

Every number here comes from B and C. None uses an A response. Appendix A2 derives the coefficient formulas.

The line’s four held-out predictions

The fitted rule is $\hat{f}_{B+C}(x) = 11 + x$. For the first A case, inserting $x = -3$ gives prediction 8. Its answer is 3, so its squared loss is $(3 - 8)^2 = 25$.

x in A	Observed y	Prediction	Residual	Squared loss
−3	3	8	−5	25
−1	9	10	−1	1
1	11	12	−1	1
3	9	14	−5	25

$$\mathrm{MSE}_A(\text{line}) = \frac{25 + 1 + 1 + 25}{4} = 13$$

On these four held-out cases, the line’s MSE of 13 improves on the mean-only score of 18.

Fitting the quadratic using B and C

At each x there are two training responses. Their mean is the best squared-error prediction for that pair.

x	B response	C response	Pair mean
−3	9	15	12
−1	7	5	6
1	9	7	8
3	15	21	18

$$\hat{f}_{B+C}(x) = 6 + x + x^2$$

This quadratic passes through all four pair means. It therefore attains the smallest possible sum for every pair at once, so it is the least-squares quadratic fit.

Appendix A3 shows how the four pair means determine the coefficients 6, 1 and 1.

The quadratic’s four held-out predictions

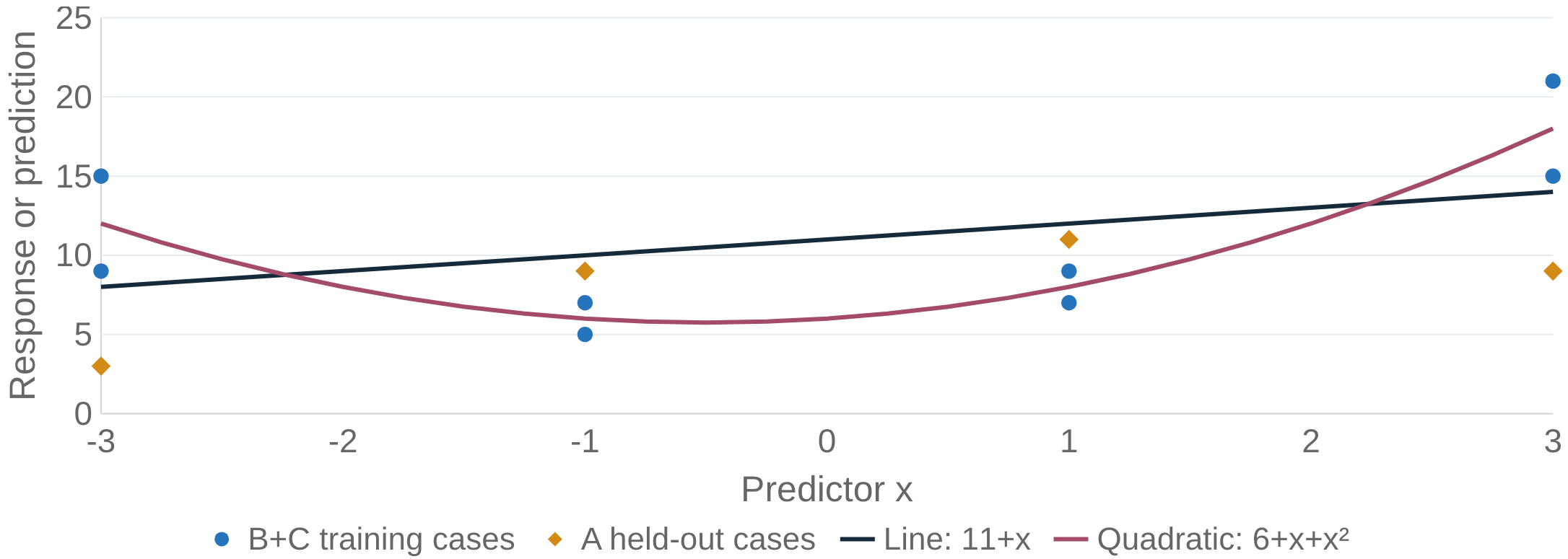
Now insert the same A inputs into $\hat{f}_{B+C}(x) = 6 + x + x^2$. For $x = -3$, the prediction is $6 - 3 + 9 = 12$.

x in A	Observed y	Prediction	Residual	Squared loss
−3	3	12	−9	81
−1	9	6	3	9
1	11	8	3	9
3	9	18	−9	81

$$\mathrm{MSE}_A(\text{quadratic}) = \frac{81 + 9 + 9 + 81}{4} = 45$$

The extra curvature helps the B+C fit, but predicts the excluded A responses poorly.

The fitted curves and the held-out cases



Both curves use only the blue training cases. The gold cases supply the validation answers. The quadratic's right-hand prediction misses A by 9 units.

Training improvement and prediction improvement

Least-squares candidate	Training MSE on B+C	Validation MSE on A
Mean only	26	18
Line	21	13
Quadratic	5	45

The quadratic gives the smallest training MSE. The line gives the smallest validation MSE on this split.

Flexibility makes it easier to fit a particular training pattern. A held-out score checks whether that fitted pattern also predicts other responses.

A particular validation score can be below its training score. Training optimism is a tendency across repeated datasets, rather than an ordering that every split must satisfy.

[1, 2]

A different split can select a different model

Suppose we hold out B instead and train on A+C. The training cases change, so we must fit every candidate again.

Candidate	Fit from A+C	Validation MSE on B
Mean only	10	9
Line	10 + x	4
Quadratic	7.5 + x + 0.5x²	0

The quadratic wins on B, whereas the line won on A. The quadratic predicts all four B answers exactly in this constructed example.

One split changes both the fitted model and the cases used to assess it. Cross-validation reduces dependence on one held-out subset by rotating that role.

The three-fold cross-validation schedule

Round	Fit on	Hold out	Training size
1	B + C	A	8
2	A + C	B	8
3	A + B	C	8

Each case supplies one held-out prediction. The other eight cases train the model that makes that prediction.

Each round begins again from the candidate specification. Reusing the previous round’s coefficients could reuse answers that are now supposed to be held out.

The folds stay the same across candidates, so the comparison uses the same assessment cases.

Round 2: the line fitted without B

$$\hat{f}_{A+C}(x) = 10 + x$$

The A+C predictor mean is 0 and response mean is 10. The slope is again 40/40 = 1, so the fitted intercept is 10.

x in B	Observed y	Prediction	Residual	Squared loss
−3	9	7	2	4
−1	7	9	−2	4
1	9	11	−2	4
3	15	13	2	4

$$\mathrm{MSE}_B(\text{line}) = \frac{4+4+4+4}{4} = 4$$

Round 3: the line fitted without C

$$\hat{f}_{A+B}(x) = 9 + x$$

The A+B predictor mean is 0 and response mean is 9. Fitting gives slope 1 and intercept 9. The C answers have no role in those estimates.

x in C	Observed y	Prediction	Residual	Squared loss
−3	15	6	9	81
−1	5	8	−3	9
1	7	10	−3	9
3	21	12	9	81

$$\mathrm{MSE}_C(\text{line}) = \frac{81 + 9 + 9 + 81}{4} = 45$$

Combining all 12 held-out prediction errors

A prediction made while its case is excluded from training is an out-of-fold prediction. We have now obtained one for every development case.

Held-out cases	Sum of squared losses	Number scored	Fold MSE
A	52	4	13
B	16	4	4
C	180	4	45
All folds	248	12	248/12

$$\mathrm{CV}_3(\text{line}) = \frac{52 + 16 + 180}{12} = \frac{62}{3} \approx 20.67$$

The denominator is 12 because we score 12 predictions. Training a case in other rounds does not add extra scored predictions for that case.

The general cross-validation formula

$$\mathrm{CV}_K = \frac{1}{n}\sum_{i=1}^{n} L\left(y_i, \hat{f}^{(-k(i))}(x_i)\right)$$

Symbol	Meaning in words	Our example
n, K	Number of cases and folds	n = 12, K = 3
k(i)	The fold containing case i	A, B or C
$\hat{f}^{(-k(i))}$	Fit with that whole fold excluded	For i in A, fit on B+C
L(y, ŷ)	Loss for one held-out prediction	$(y - \hat{y})^2$

The formula is the same calculation as the previous table: obtain each case’s excluded-fold prediction, calculate its loss, then average over cases.

Classification uses the same exclusion rule

For classification, replace squared loss with 0–1 loss: a correct predicted label contributes 0 and an incorrect label contributes 1.

Case	1	2	3	4	5	6
Actual label	Yes	No	Yes	No	No	Yes
Out-of-fold label	Yes	Yes	Yes	No	No	No
0–1 loss	0	1	0	0	0	1

$$\text{CV error rate} = \frac{0+1+0+0+0+1}{6} = \frac{1}{3}$$

Each label prediction must come from training without that case’s fold. The exclusion logic stays the same even though the loss changes.

Averaging folds when sizes differ

Our folds have four cases each, so averaging their MSEs gives the same result as averaging all 12 losses.

$$\frac{13+4+45}{3} = \frac{4(13)+4(4)+4(45)}{12} = \frac{62}{3}$$

If folds differ in size, the pooled per-case score weights each fold by its number of cases:

Fold	Number of cases	Fold MSE	Squared-loss sum
1	2	1	2 × 1 = 2
2	3	4	3 × 4 = 12

$$\text{CV} = \frac{2+12}{5} = 2.8 \qquad \text{rather than} \qquad \frac{1+4}{2} = 2.5$$

The value 2.5 gives equal weight to folds. The value 2.8 gives equal weight to cases, which is the score defined here.

Comparing the candidates using all folds

Candidate	MSE on A	MSE on B	MSE on C	3-fold CV MSE
Mean only	18	9	50	77/3 ≈ 25.67
Line	13	4	45	62/3 ≈ 20.67
Quadratic	45	0	45	30

The CV score chooses the line from these three candidates. It predicts better on average across the three held-out subsets.

Each quadratic is also refitted in its round: $6 + x + x^2$, then $7.5 + x + 0.5x^2$, then $9 + x$. A zero quadratic coefficient is an allowed fit.

This is a conclusion about these data and candidates. A different dataset may support a different degree.

The final fit uses all development cases

Cross-validation has selected the line family. We now estimate its coefficients again, using all 12 development cases.

$$\bar{x} = 0, \quad \bar{y} = 10, \quad \sum x_i^2 = 60, \quad \sum x_i y_i = 60$$

$$\hat{b} = 1, \qquad \hat{a} = 10, \qquad \hat{f}_{\text{final}}(x) = 10 + x$$

This final fit is a fourth line fit. The earlier three fits existed to generate assessment predictions.

We do not deploy the fold with the lowest error or average its fitted coefficients as part of ordinary CV. We refit the chosen specification on all available development data.

The test responses are still untouched. They will assess this chosen and refitted rule later.

What the CV score estimates

Our three-fold score uses models trained on eight cases. The final line uses twelve. Their prediction errors need not be identical.

Cross-validation approximates the prediction performance of the specified training procedure at the fold training size.

If the training data changed, the fitted coefficients and CV score could change. A score from one dataset is an estimate, not a known population error.

Here the same CV scores also selected the model family. The winning score can benefit from selection, a separate issue addressed after LOOCV.

Appendix B states the expected-error targets more precisely. Normality of the errors is unnecessary for defining CV.

[1, 8]

The number of folds and the training size

Smaller held-out folds leave more data for each fit. More folds also require more fits for each candidate.

K for n = 12	Held out per fit	Train per fit	Fits per candidate
2	6	6	2
3	4	8	3
6	2	10	6
12	1	11	12

For larger datasets, 5-fold and 10-fold CV are common choices. The best choice depends on the data, procedure and computation.

At K = n, each round holds out one case. This is leave-one-out cross-validation, abbreviated LOOCV.

[1, 2]

LOOCV with three observations

Consider a separate three-case example and a fixed straight-line family. Holding out one case leaves two points, which determine its fitted line.

Omitted case (x, y)	Line fitted on other two	Prediction	Squared loss
(−1, 1)	$\hat{f}^{(-1)}(x) = x$	−1	4
(0, 0)	$\hat{f}^{(-2)}(x) = 1$	1	1
(1, 1)	$\hat{f}^{(-3)}(x) = -x$	−1	4

$$\text{LOOCV} = \frac{4 + 1 + 4}{3} = 3$$

Every prediction comes from a fit that excludes the response being scored. We will recover this same value from one full-data OLS fit.

Ordinary residuals and omitted-case residuals

$$e_i = y_i - \hat{f}(x_i), \qquad e_i^{(-i)} = y_i - \hat{f}^{(-i)}(x_i)$$

The ordinary residual e_i uses a model fitted to all cases, including the i-th answer. The omitted-case residual $e_i^{(-i)}$ uses a model fitted without that answer.

For the three-case example, symmetry gives slope 0 and the response mean gives intercept 2/3. Thus $\hat{f}(x) = 2/3$ and its ordinary residuals are 1/3, −2/3 and 1/3.

$$\text{Training MSE} = \frac{(1/3)^2 + (-2/3)^2 + (1/3)^2}{3} = \frac{2}{9}$$

This is much smaller than LOOCV = 3. To recover omitted-case errors from the full fit, we need to account for each answer's contribution to its own fitted value.

Choosing the smallest score is another learning step

We used the CV scores to choose the line. That choice also learned from the development answers. We now examine how this changes the meaning of the winning score.

Hypothetical candidate	True future MSE	Observed CV score
A	10	12
B	10	8
C	10	11

All three true errors are equal in this illustration. Choosing B because it scored 8 also chooses the candidate that received the most favourable scoring fluctuation.

Assessment after model choice

The selected minimum can look better than the selected procedure’s future error. A favourable scoring fluctuation may have helped the candidate win.

The winning CV score has already helped choose the model. A separate test set can assess the result using responses that played no role in that choice.

In our example, every line score excluded its assessment answers from coefficient fitting. But the collection of scores used all development answers to choose the line family.

That distinction explains why valid out-of-fold predictions are useful for selection while the selected minimum still needs a fresh performance assessment.

The numbers above illustrate the mechanism. Appendix D gives an expectation argument for a fixed candidate set.

[5, 6]

The untouched test set assesses the final rule

We chose the line using development CV and refitted it as $\hat{f}_{\text{final}}(x) = 10 + x$. Now, for the first time, use the four reserved test answers.

Test x	Test y	Final prediction	Residual	Squared loss
−2	9	8	1	1
0	9	10	−1	1
2	14	12	2	4
3	12	13	−1	1

$$\text{Test MSE} = \frac{1 + 1 + 4 + 1}{4} = 1.75$$

CV and test score different cases using fits trained on 8 and 12 cases. Their scores need not agree. Four constructed test cases give an imprecise assessment.

Using the test results to change degree would make them part of model choice. Further assessment would then need fresh held-out data.

Preprocessing also learns from data

A prediction procedure can learn more than regression coefficients. It may learn missing-value replacements, standardization constants or a feature list.

For each validation round, learn these quantities from that round's training data and then apply the fitted transformations to the held-out inputs.

For example, standardization subtracts a training mean and divides by a training scale. Computing that mean on all cases lets held-out inputs influence the fitted procedure.

$$z = \frac{x - \mu_T}{s_T} \qquad \text{with } \mu_T, s_T \text{ learned using training inputs only}$$

A fixed conversion such as centimetres to metres learns no quantity from the data. The training-only rule concerns steps that estimate something.

[7]

A numerical preprocessing example

Suppose one round’s training inputs are 0 and 2. Define the scale here using the mean squared deviation, with divisor 2.

$$\mu_T = \frac{0+2}{2} = 1, \qquad s_T = \sqrt{\frac{(0-1)^2 + (2-1)^2}{2}} = 1$$

Input	Transformation using training values	Standardized value
Training x = 0	(0 − 1) / 1	−1
Training x = 2	(2 − 1) / 1	1
Held-out x = 4	(4 − 1) / 1	3

Fit the model on the transformed training cases. Transform x = 4 with the same mean and scale before predicting its answer.

Do not recompute the constants using 0, 2 and 4. Some models are invariant to a change of scale, but the training-only protocol must cover the full procedure.

Feature selection can leak the held-out answers

Imagine many irrelevant predictors and a response unrelated to all of them. By chance, a few predictors may appear correlated with the full set of responses.

If we select those predictors using all responses and only then cross-validate the fitted model, the held-out answers have already influenced which inputs the model receives.

In a correct round:
1. Use the training responses to choose the feature list.
2. Fit the model using those training cases and selected features.
3. Use that fixed list to predict the held-out cases.

The chosen features can differ across folds. Cross-validation should reproduce every learned step of the procedure we plan to use.

[5, 7]

Nested CV when a separate test set is unavailable

Ordinary CV can choose a setting. To assess the procedure that makes this choice, an outer loop must keep some cases outside the entire selection process.

One outer round with 12 development cases:

1. Put four outer held-out cases aside.
2. On the remaining eight, use inner CV to compare candidates.
3. Choose a candidate using only those inner scores.
4. Refit it on all eight outer training cases.
5. Predict and score the four outer held-out cases.

All preprocessing and feature selection also take place inside the appropriate training portions.

The outer answers assess the result of choosing and fitting. They do not choose the candidate in that round.

[5, 6]

What the outer loop returns

Outer round	Data used to choose and fit	Data used only to score
1	B+C, including inner CV	A
2	A+C, including inner CV	B
3	A+B, including inner CV	C

Different outer rounds may choose different candidates. The outer score assesses the procedure that selects a candidate and fits it.

Average the outer held-out losses over cases. This evaluates that procedure at the outer training size, eight cases here.

To build the final model, rerun the selection procedure on all development data and refit the chosen candidate there. Choosing settings from outer scores would reuse the assessment data.

[5, 6]

A1 Why least squares predicts the response mean

Consider a constant prediction a for m training responses. Let $\bar{y}$ be their mean. Add and subtract $\bar{y}$ inside each squared residual.

$$\sum_{i=1}^{m}(y_i-a)^2=\sum_{i=1}^{m}[(y_i-\bar{y})+(\bar{y}-a)]^2$$

$$=\sum_{i=1}^{m}(y_i-\bar{y})^2+2(\bar{y}-a)\sum_{i=1}^{m}(y_i-\bar{y})+m(\bar{y}-a)^2$$

The middle sum is zero because deviations from the mean sum to zero. The first term does not depend on a. The last term is nonnegative and equals zero at $a=\bar{y}$.

Thus $a=\bar{y}$ minimizes the training sum. With B+C, the eight responses sum to 88, so the fitted constant is 11.

A2 The least-squares line coefficients

$$S(a,b) = \sum_{i=1}^{m} (y_i - a - bx_i)^2$$

At a minimum, both first derivatives vanish. Setting the derivative with respect to a to zero gives:

$$\sum_i (y_i - a - bx_i) = 0 \qquad \Longrightarrow \qquad a = \bar{y} - b\bar{x}$$

Substitute this into the slope equation. Centring x and y removes the intercept:

$$\sum_i (x_i - \bar{x})[(y_i - \bar{y}) - b(x_i - \bar{x})] = 0$$

$$\hat{b} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{a} = \bar{y} - \hat{b}\bar{x}$$

A unique slope requires at least two distinct predictor values. B+C has $\bar{x} = 0$, so the two slope sums simplify to $\Sigma x_i y_i$ and Σx_i^2.

A3 The quadratic fit in the worked example

For two responses u and v at the same x, any prediction q satisfies:

$$(u-q)^2+(v-q)^2=\frac{(u-v)^2}{2}+2\left(q-\frac{u+v}{2}\right)^2$$

The pair's smallest sum occurs at q = (u+v)/2. For B+C, these means at x = −3, −1, 1, 3 are 12, 6, 8, 18.

$$a-b+c=6,\quad a+b+c=8 \implies b=1,\ a+c=7$$

$$a-3b+9c=12 \implies a+9c=15$$

$$8c=8,\quad c=1,\quad a=6 \implies \hat{f}(x)=6+x+x^2$$

At x = 3, this also gives 18. The quadratic attains every pair's minimum simultaneously, proving it minimizes the full training sum.

B1 Prediction risk for a fixed fitted rule

Let T denote a training dataset and $\hat{f}_T$ the rule fitted using it. Let (Xnew, Ynew) be an independent new case from the target population.

$$R(\hat{f}_T) = \mathbb{E}_{\text{new}}\left[L\left(Y_{\text{new}}, \hat{f}_T(X_{\text{new}})\right) \mid T\right]$$

This expectation averages the prediction loss over possible future cases while holding the observed training data and fitted rule fixed.

$$\hat{R}_V = \frac{1}{|V|}\sum_{i \in V} L(y_i, \hat{f}_T(x_i)), \qquad \mathbb{E}[\hat{R}_V \mid T] = R(\hat{f}_T)$$

The equality holds when V contains independent target-population cases and the rule is fixed before looking at V. A finite validation score still has sampling variability.

[1, 8]

B2 The training-procedure target of CV

A procedure maps a training sample to a fitted rule. Its expected prediction risk at training size m averages over possible training samples as well as future cases.

$$R_m = \mathbb{E}_{T_m}\left[R(\hat{f}_{T_m})\right]$$

With independent identically distributed cases, folds fixed independently of their values, and a candidate procedure fixed in advance, each held-out loss evaluates a fit based on the other cases.

For equal folds, the training size is m = n(1 − 1/K). Thus the corresponding CV score targets Rm, rather than the exact risk of the final fit on all n observed cases.

Using CV to choose the candidate adds selection. Grouped or time-dependent data require a target and split appropriate to their dependence.

[1, 8]

D1 Why the selected minimum is optimistic

Suppose J fixed candidate rules have true risks $R_1,\ldots,R_J$. An independent validation set gives estimates $\hat{R}_j$ with $E[\hat{R}_j] = R_j$. Let $\hat{j}$ choose the smallest observed estimate.

$$\hat{j} = \operatorname*{argmin}_j \hat{R}_j, \qquad \hat{R}_{\hat{j}} = \min_j \hat{R}_j$$

The minimum is no larger than any one estimate. Taking expectations gives the first inequality below. The selected candidate's true risk is at least the best candidate risk, giving the second.

$$\mathbb{E}[\hat{R}_{\hat{j}}] \;\le\; \min_j \mathbb{E}[\hat{R}_j] = \min_j R_j \;\le\; \mathbb{E}[R_{\hat{j}}]$$

Thus the score used to select the winner is optimistic in expectation for the winner's risk under these assumptions. Independence between candidates' scoring errors is unnecessary.

This argument explains selection bias. Actual CV also has the training-size and dependence issues described earlier.

[5, 6]

References: foundations and LOOCV

[1] Hastie, Tibshirani and Friedman (2009). The Elements of Statistical Learning, 2nd edition, Chapter 7. Prediction error, model assessment and cross-validation.

[2] James, Witten, Hastie and Tibshirani (2021). An Introduction to Statistical Learning, 2nd edition, Section 5.1. Validation-set, leave-one-out and K-fold methods.

[3] Wasserman (2008). Statistical Machine Learning, Homework 2 solutions, Question 7. Algebra of the OLS leave-one-out identity.

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Links in this reference list open the source. All numerical datasets in the tutorial are constructed teaching examples.

References: selection and practical validation

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